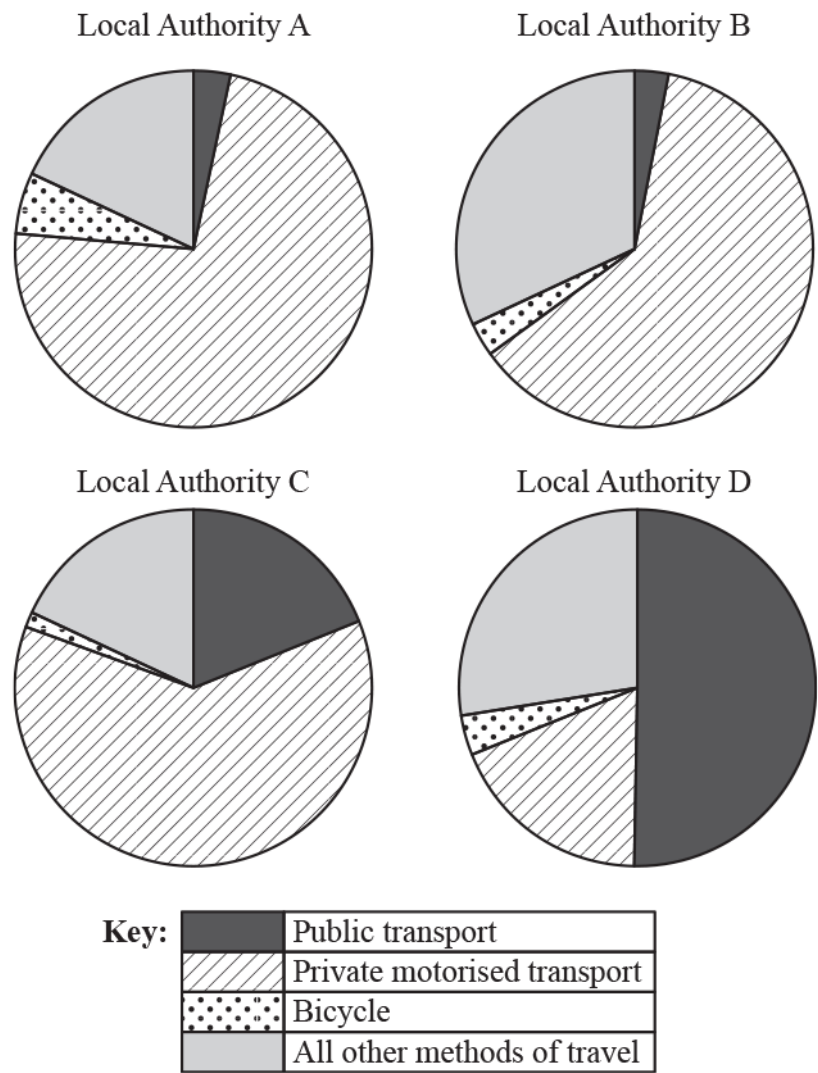


1(a) The four pie charts illustrate the numbers of employees using different methods of travel in four Local Authorities in 2011.



State, with reasons, which of the four Local Authorities is most likely to be a rural area with many hills. [2]

(b) Explain why pie charts are more suitable for answering part (a) than bar charts showing the same data. [1]

(c) Two of the Local Authorities represent urban areas.

(i) State with a reason which **two** Local Authorities are likely to be urban.

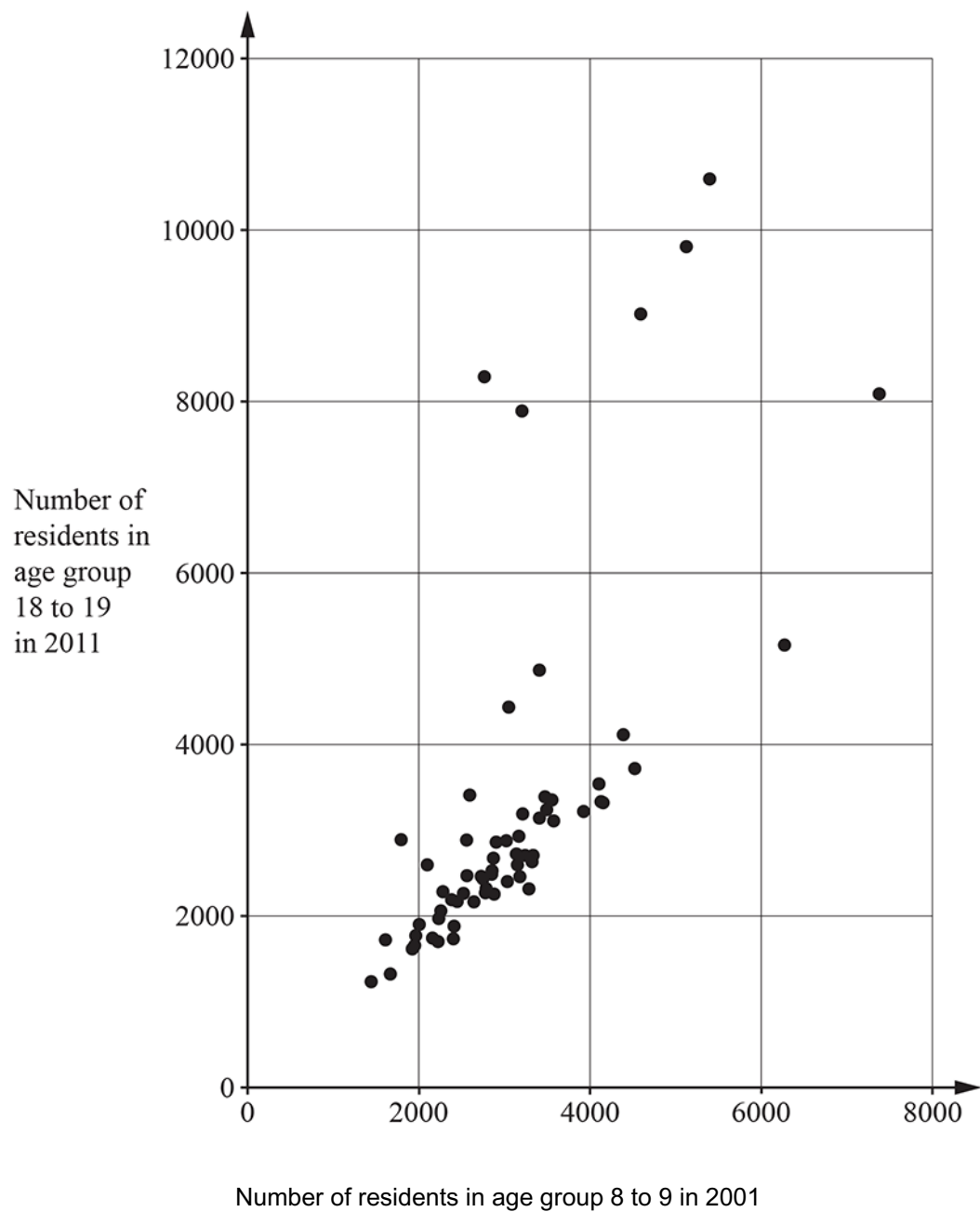
[2]

(ii) One urban Local Authority introduced a Park-and-Ride service in 2006. Users of this service drive to the edge of the urban area and then use buses to take them into the centre of the area. A student claims that a comparison of the corresponding pie charts for 2001 (not shown) and 2011 would enable them to identify which Local Authority this was.

State with a reason whether you agree with the student.

[2]

2(a) A student is investigating changes in the number of residents in Local Authorities in the South-East Region between 2001 and 2011. The scatter diagram shows the number x of residents in these Local Authorities in the age group 8 to 9 in 2001 and the number y of residents in the same Local Authorities in the age group 18 to 19 in 2011.



Suggest a reason why the student is comparing these two age groups in 2001 and 2011.

[1]

(b) The student notices that most of the data points are close to the line $y = x$.

(i) Explain what this suggests about the residents in these Local Authorities.

[1]

(ii) The student says that correlation does not imply causation, so there is no causal link between the values of x and the values of y . Explain whether or not they are correct.

[1]

(c) Some of these Local Authorities contain universities.

(i) On the diagram, circle three points that are likely to represent Local Authorities containing universities.

[1]

(ii) Give a reason for your choice of points in part (i).

[1]

(d) Assume that the proportion of residents in age group 8 to 9 in 2001 was roughly the same in each Local Authority in the South-East. The Local Authority in this region with the largest population is Medway. On the diagram, label clearly with the letter M the point that corresponds to Medway.

[1]

3(a) Table 1 shows the numbers of usual residents in the age range 0 to 4 in 15 Local Authorities (LAs) in 2001 and 2011. The table also shows the increase in the numbers in this age group, and the same increase as a percentage.

	2001	2011	Increase	% Increase
Bolton	16 779	18 765	1 986	11.84%
Bury	11 117	12 235	1 118	10.06%
Knowsley	9 454	9 121	-333	-3.52%
Liverpool	24 840	26 099	1 259	5.07%
Manchester	24 693	36 413	11 720	47.46%
Oldham	15 196	16 491	1 295	8.52%
Rochdale	13 771	14 754	983	7.14%
Salford	12 529	16 255	3 726	29.74%
Sefton	14 896	14 601	-295	-1.98%
St. Helens	10 083	10 269	186	1.84%
Stockport	16 457	17 342	885	5.38%
Tameside	12 803	14 439	1 636	12.78%
Trafford	11 971	14 870	2 899	24.22%
Wigan	17 561	19 681	2 120	12.07%
Wirral	17 475	18 514	1 039	5.95%

Table 1

Fig. 2 shows the increase in each LA in raw numbers, and Fig. 3 shows the percentage increase in each LA.

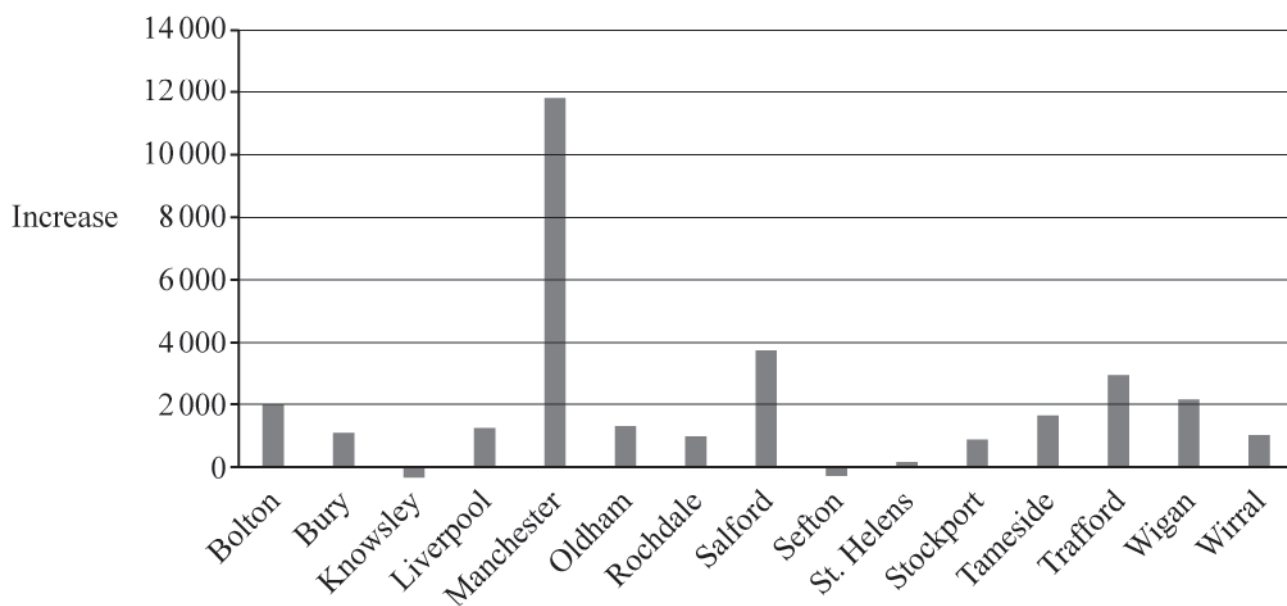


Fig. 2

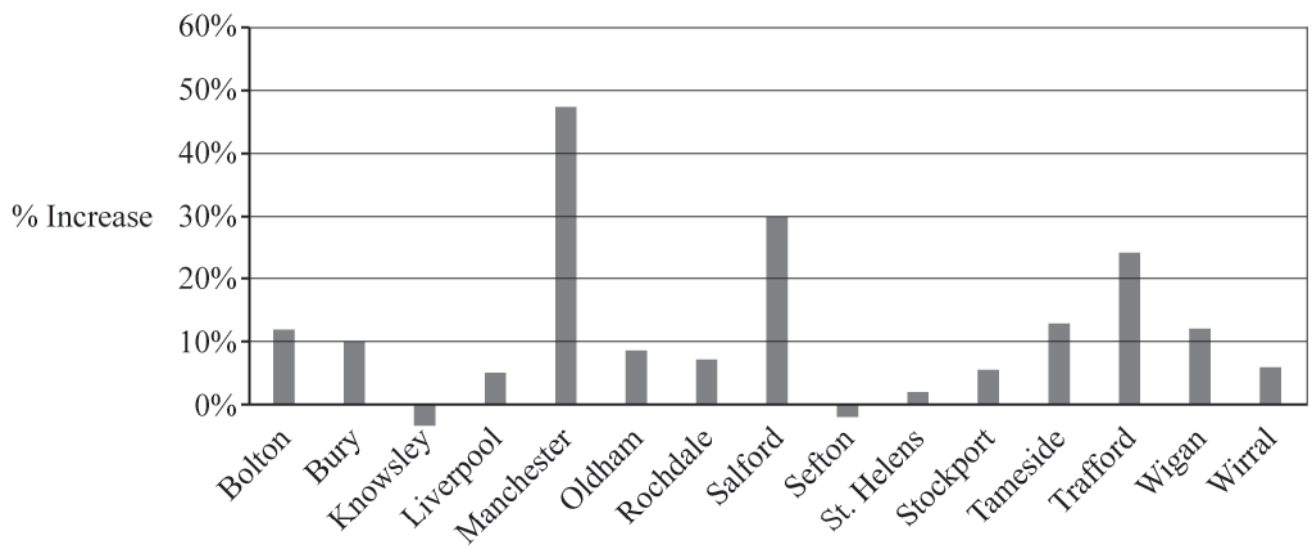


Fig. 3

In two of the 15 LAs the proportion of young families is greater than in the other 13 LAs. Suggest, using only data from Fig. 2 and Fig. 3 and / or Table 1, which two LAs these are most likely to be.

[2]

(b) The Education Committees in these LAs need to plan for the provision of schools for pupils in their districts.

(i) Explain why, in this context, the increase is more important than the actual numbers. [1]

(ii) In which of the following LAs was there likely to have been the greatest need for extra teachers in the years following 2011: Bolton, Sefton, Tameside or Wigan?
Give a reason for your answer. [2]

(iii) State an assumption about the populations needed to make your answer in part (ii) valid. [1]

4(a) A researcher is studying changes in behaviour in travelling to work by people who live outside London, between 2001 and 2011.

He chooses the 15 Local Authorities (LAs) outside London with the largest decreases in the percentage of people driving to work, and arranges these in descending order.

The table shows the changes in percentages from 2001 to 2011 in various travel categories, for these Local Authorities.

Local Authority	Work mainly at or from home	Under-ground, metro, light rail, tram	Train	Bus, minibus or coach	Driving a car or van	Passenger in a car or van	Bicycle	On foot
Brighton and Hove	3.2	0.1	1.5	0.8	-8.2	-1.5	2.1	2.3
Cambridge	2.2	0.0	1.6	1.2	-7.4	-1.0	3.1	0.6
Elmbridge	2.9	0.4	4.1	0.2	-6.6	-0.7	0.3	-0.3
Oxford	2.0	0.0	0.6	-0.4	-5.2	-1.1	2.2	2.1
Epsom and Ewell	1.6	0.4	3.9	1.1	-5.2	-0.9	0.0	-0.6
Watford	0.7	2.0	3.1	0.4	-4.5	-1.2	0.0	-0.1
Tandridge	3.3	0.2	4.0	-0.1	-4.5	-1.1	0.0	-1.3
Mole Valley	3.3	0.1	1.9	0.3	-4.4	-0.7	0.2	-0.3
St Albans	2.3	0.3	3.4	-0.3	-4.3	-1.2	0.3	-0.2
Chiltern	2.9	1.4	1.4	0.1	-4.2	-0.6	-0.2	-0.8
Exeter	0.7	0.0	1.0	-0.6	-4.2	-1.5	1.7	3.4
Woking	2.1	0.1	3.7	0.0	-4.2	-1.3	-0.1	0.0
Reigate and Banstead	1.8	0.1	3.2	0.6	-4.1	-1.0	0.1	-0.2
Waverley	4.3	0.1	2.5	-0.5	-3.9	-0.9	-0.3	-0.9
Guildford	2.7	0.1	2.4	0.2	-3.6	-1.2	0.0	-0.3

Explain why these LAs are not necessarily the 15 LAs with the largest decreases in the percentage of people driving to work.

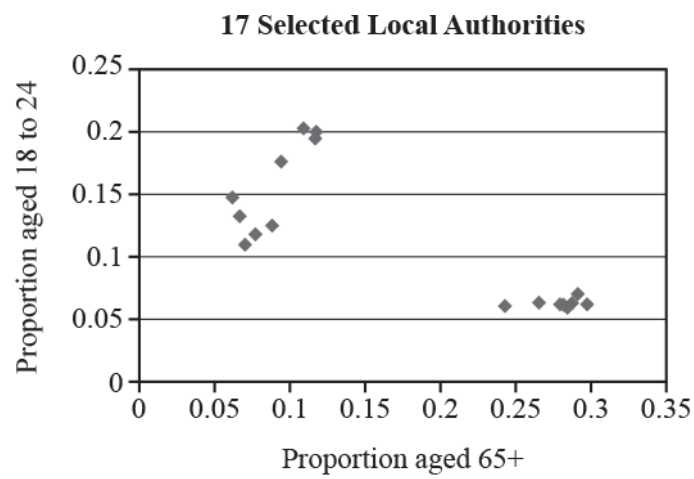
[1]

- (b) The researcher wants to talk to those LAs outside London which have been most successful in encouraging people to change to cycling or walking to work.
Suggest four LAs that he should talk to and why.

[2]

- (c) Which two categories have replaced driving to work for the highest percentages of workers in these LAs?
Support your answer with evidence from the table. [3]

5(a) A trainer was asked to give a lecture on population profiles in different Local Authorities (LAs) in the UK. Using data from the 2011 census, he created the following scatter diagram for 17 selected LAs.



He selected the 17 LAs using the following method. The proportions of people aged 18 to 24 and aged 65+ in any Local Authority are denoted by P_{young} and P_{senior} respectively. The trainer used a spreadsheet to calculate the

value of $k = \frac{P_{\text{young}}}{P_{\text{senior}}}$ for each of the 348 LAs in the UK. He then used specific ranges of values of k to select the 17 LAs.

Estimate the ranges of values of k that he used to select these 17 LAs.

[2]

(b) Using the 17 LAs the trainer carried out a hypothesis test with the following hypotheses.

H_0 : There is no linear correlation in the population between P_{young} and P_{senior} .

H_1 : There is negative linear correlation in the population between P_{young} and P_{senior} .

He found that the value of Pearson's product-moment correlation coefficient for the 17 LAs is - 0.797, correct to 3 significant figures.

(i) Use the Critical Values table in part d) below to show that this value is significant at the 1% level. [2]

(ii) The trainer concluded that there is evidence of negative linear correlation between P_{young} and P_{senior} in the population.

Use the scatter diagram from part a) to comment on the reliability of this conclusion. [2]

(c) Describe one outstanding feature of the population in the areas represented by the points in the bottom right hand corner of the diagram. [1]

(d) The trainer's audience included representatives from several universities.

Suggest a reason why the diagram might be of particular interest to these people.

[1]

Critical values of Pearson's product-moment correlation coefficient

1-tail test	5%	2.5%	1%	0.5%
2-tail test	10%	5%	2%	1%
<i>n</i>				
1	-	-	-	-
2	-	-	-	-
3	0.9877	0.9969	0.9995	0.9999
4	0.9000	0.9500	0.9800	0.9900
5	0.8054	0.8783	0.9343	0.9587
6	0.7293	0.8114	0.8822	0.9172
7	0.6694	0.7545	0.8329	0.8745
8	0.6215	0.7067	0.7887	0.8343
9	0.5822	0.6664	0.7498	0.7977
10	0.5494	0.6319	0.7155	0.7646
11	0.5214	0.6021	0.6851	0.7348
12	0.4973	0.5760	0.6581	0.7079
13	0.4762	0.5529	0.6339	0.6835
14	0.4575	0.5324	0.6120	0.6614
15	0.4409	0.5140	0.5923	0.6411
16	0.4259	0.4973	0.5742	0.6226
17	0.4124	0.4821	0.5577	0.6055
18	0.4000	0.4683	0.5425	0.5897
19	0.3887	0.4555	0.5285	0.5751
20	0.3783	0.4438	0.5155	0.5614
21	0.3687	0.4329	0.5034	0.5487
22	0.3598	0.4227	0.4921	0.5368
23	0.3515	0.4132	0.4815	0.5256
24	0.3438	0.4044	0.4716	0.5151
25	0.3365	0.3961	0.4622	0.5052
26	0.3297	0.3882	0.4534	0.4958
27	0.3233	0.3809	0.4451	0.4869
28	0.3172	0.3739	0.4372	0.4785
29	0.3115	0.3673	0.4297	0.4705
30	0.3061	0.3610	0.4226	0.4629

6(a) The table shows extracts from the “Method of travel by LA” tabs for 2001 and 2011 in the large data set.

Local authority (LA)	All people in employment	Underground, metro, light rail, tram	Train	Bus, minibus or coach	Motorcycle, scooter or moped	Driving a car or van
LA1 2001	79 226	14 369	5235	20 575	1227	16 052
LA1 2011	118 556	22 486	8336	30 541	1220	12 445
LA2 2001	203 614	190	1062	15 327	1256	121 690
LA2 2011	227 894	323	1865	13 732	1038	146 644
LA3 2001	42 993	35	482	4363	274	24 105
LA3 2011	49 014	33	828	3380	191	28 981
LA4 2001	101 697	65	693	21 758	846	45 407
LA4 2011	123 218	2495	1315	24 275	763	54 020

In one of these four LAs a new tram system was opened in 2004.

Suggest, with a reason taken from the data, which LA this could have been.

[2]

- (b) Julian suggests that the figures for “Bus, minibus or coach” for LA1 show that some new bus routes were probably introduced in this LA between 2001 and 2011.

Use data from the table to comment on this suggestion.

[2]

- (c) In one of these four LAs a congestion charge on vehicles was introduced in 2003.

Suggest, with a reason taken from the data, which LA this could have been.

[2]

- 7 Using the 2001 UK census results and some software, Javid intended to calculate the mean number of people who travelled to work by underground, metro, light rail or tram (UMLT) for all 348 Local Authorities. However, Javid noticed that for one LA the entry in the UMLT column is a dash, rather than a 0. See the extract below.

Data extract for one LA in 2001			
Work mainly at or from home	UMLT	Train	Bus, minibus or coach
295	–	4	4

Javid felt that it was not clear how this LA was to be treated so he decided to omit it from his calculation.

- (a) Explain how the omission of this LA affects Javid's calculation of the mean. [1]

The value of the mean that Javid obtained was 2046.3.

- (b) Calculate the value of the mean when this LA is not removed. [2]

Javid finds that the corresponding mean for all Local Authorities for 2011 is 2860.8. In order to compare the means for the two years, Javid also finds the total number of employees in each of these years. His results are given below.

Year	2001	2011
Total number of employees	23 627 753	26 526 336

(c) Show that a higher proportion of employees used the metro to travel to work in 2011 than in 2001.

[2]

(d) Suggest a reason for this increase.

[1]

- 8 The table shows information, derived from the 2011 UK census, about the percentage of employees who used various methods of travel to work in four Local Authorities.

Local Authority	Underground, metro, light rail or tram	Train	Bus	Drive	Walk or cycle
A	0.3%	4.5%	17%	52.8%	11%
B	0.2%	1.7%	1.7%	63.4%	11%
C	35.2%	3.0%	12%	11.7%	16%
D	8.9%	1.4%	9%	54.7%	10%

One of the Local Authorities is a London borough and two are metropolitan boroughs, not in London.

- (a) Which one of the Local Authorities is a London borough? Give a reason for your answer. [1]

- (b) Which two of the Local Authorities are metropolitan boroughs outside London? In each case give a reason for your answer. [2]

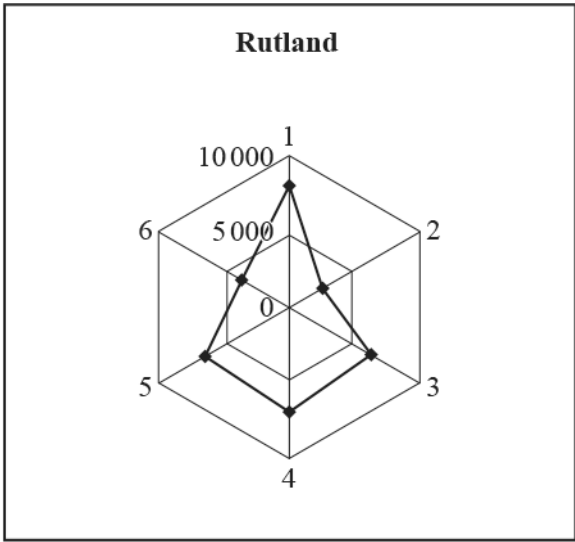
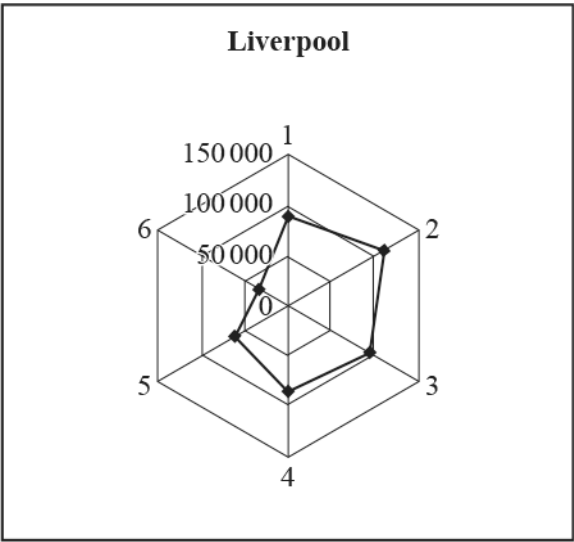
(c) Describe one difference between the public transport available in the two metropolitan boroughs, as suggested by the table.

[1]

(d) Comment on the availability of public transport in Local Authority B as suggested by the table.

[2]

9 The radar diagrams illustrate some population figures from the 2011 census results.



Each radius represents an age group, as follows:

Radius	1	2	3	4	5	6
Age group	0–17	18–29	30–44	45–59	60–74	75+

The distance of each dot from the centre represents the number of people in the relevant age group.

- (a) The scales on the two diagrams are different. State an advantage and a disadvantage of using different scales in order to make comparisons between the ages of people in these two Local Authorities. [2]

Advantage:

Disadvantage:

(b) Approximately how many people aged 45 to 59 were there in Liverpool?

[1]

(c) State the main two differences between the age profiles of the two Local Authorities.

[2]

(d) James makes the following claim.

“Assuming that there are no significant movements of population either into or out of the two regions, the 2021 census results are likely to show an increase in the number of children in Liverpool and a decrease in the number of children in Rutland.”

Use the radar diagrams to give a justification for this claim.

[2]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1	a		A: High private and low public OR B: Low bicycle and low public OR B: Low bicycle and high private OR C: Low bicycle and high private	B2 [2]	Allow B1 for A or B or C with one correct factor only. Ignore else All answers can be implied Ignore all else <u>Examiner's Comments</u> Because questions of this kind generally involve inference, rather than deduction, there is not always one single, precisely correct answer to any particular question. This is a result of the increased emphasis, in the new Statistics specification, on interpretation as compared to calculation. Various answers were acceptable in each part of this question. Centres are referred to the published mark scheme for details of these.
	b		Pie charts allow comparison of proportions Pie charts show proportions of Ignore all else	B1 [1]	NOT: Bar charts don't show proportions unless also state pies do. Assume "They" means pie charts. NOT: It's easier to compare data Pie charts don't easily show which is greatest Allow percentages instead of proportions
	c	i	C and D Larger (or high or most) proportion public transport Can be implied Ignore all else	B1 B1 [2]	Both, no others Dep applied to at least one of C and D, even if other LA mentioned B-marks are independent

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	LDS says “The method of travel used is for the longest part, by distance, of the usual journey to work” Correct answers therefore should assume users of P&R will report only private, not public Method of travel is the type used for the longest part of the journey Most people using Park-and-Ride would still say they were using private transport. People will still have to travel by car, so the results won't be very different Alternatives, assuming (incorrectly) users of P&R will report both methods or just public: Public transport increase or private decrease P&R will result in more people using private transport, so this will show an increase Users of P&R will use both public and private, so not clear No, because other changes might have been made that affect the proportions If P&R users report both, or just public, then yes, public will show increase Recognition of issue as to whether P&R users should report private or public Hence unclear whether change will show	B1 B1 B2 Alternatives, assuming (incorrectly) users of P&R will report both methods or just public: B1 B0 B1 B0 B1 B0 B1 B0 B2 B2 B1 B1 [2]	“Disagree” may be implied independent Ignore all else Alternatives, assuming (incorrectly) users of P&R will report both methods or just public: “Agree” may be implied Ignore all else Not just “Increase in private” without justification Other sensible answers may be seen not covered in this MS <u>Examiner's Comments</u>  Misconception Most candidates appeared to be unfamiliar with the comment in row 21 of the Information Sheet at the beginning of the Large Data Set: “The method of travel used is for the longest part, by distance, of the usual journey to work.” They assumed that employees using the Park and Ride would report both methods of travel in the census. It would be helpful for centres to make candidates aware of the Information Sheet, and particularly row 21 and the terminology in rows 8 to 12.
			Total	7	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
2	a		Those in age group 8 to 9 in 2001 will be in age group 18 to 19 in 2011	B1 [1]	or, eg, "Same people" May be implied
	b	i	Few people joined (or left) this group oe or "Not much change in this group" or "Most have lived there since 2001"	B1 [1]	LAs with a high number of 8-9s in 2001 have a high number of 18-19s in 2011 Most have stayed in the same area since childhood
		ii	(No) The nos of 18-19's in 2011 is (largely) caused by the nos of 8-9s in 2001, (but this cannot be inferred from the correlation).	B1 [1]	(No) There is a link. They are the same people Must state or imply that there is a link between x and y in this context.
	c	i	3 (or 4 or 5) of the highest five y-values	B1 [1]	Must not include right-most point
		ii	In these LAs, the no. of 18-19s in 2011 is more than the no. of 8-9s in 2001 In these LAs there are large numbers of 18+	B1 [1]	eg, the no. of people in this group in 2011 is more than it was in 2001 Students are aged 18+ so they cause an increase in this age group Allow numerical example. May be implied Not just "These are outliers" oe
	d		Right most point indicated	B1 [1]	
			Total	6	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
3	a		Manchester and Salford Highest % or absolute increase Manchester and Liverpool The two highest in 2011 SC Manchester (alone), Highest % or absolute increase:	B1 B1 dep B1 B1 dep [2]	Wigan and Bolton Highest numbers in 2011 except Manchester and Liverpool, which are very large Salford and Trafford They have the largest absolute (or %) increase, but are small (or not huge like Manchester) B1 B1 dep B1 B1 dep
	b	i	The actual number of extra pupils determines the number of places needed Shows how many new students there will be Shows trend so LA can provide accordingly Need to know expected number of pupils	B1 (AO2.2b) [1]	The existing numbers are already catered for Increase in provision Not Need to know increase in proportion of pupils
		ii	Wigan Increase in number is greatest there	B1 (AO2.2b) B1 (AO2.2b) [2]	Allow "Wigan and Bolton" Ignore mention of % increase. Ignore extras.
		iii	E.g. all those in this category stay in the LA Populations continue growing at same rate Populations all growing at same rate The population increases consistently NOT Increase has been steady All LAs have the same teacher / pupil ratio All LAs have same need for teachers in 2011	B1 (AO2.2b) [1]	No decrease in population Children born in that LA will go to school in that LA Assume no great influx or outflow of children after 2011. The LAs are not currently understaffed Ignore extra eg "between 2001 & 2011"
			Total	6	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
4	a		The table does not include London LAs, so one of them might have had a greater increase. These areas may not have as large decrease as areas such as London	B1(AO2.2b) [1]	NOT Not include London plus incorrect, eg "small sample" Not include London and other cities There could be other LAs with bigger decreases in other years
	b		Brighton and Hove, Oxford, Cambridge, Exeter. E.g. they have relatively high (positive) values in the Bicycle and Walking columns	B1*(AO1.1) B1dep(AO2.2a) [2]	All four required or, eg, they have the largest increases (or changes) in percentage cycling & walking or they have largest total increase for cycling and walking Must mention both cycling and walking
	c		Work mainly at or from home Train Eg: These two categories have entries above 1.0 except in 3 cases. These two categories' entries all above 0.6 No other method of travel has more than 3 entries greater than 2.0 Both are all positive <u>and</u> have largest home (4.3) and largest train (4.1) Both are all over 0.5. Smallest home 0.7, smallest train 0.6 Average increases: Home 2.4, Train 2.55 (must be correct) Total increases: Home 36, Train 38.3 (correct)	B1(AO2.2b) B1(AO2.2b) B1(AO2.4) [3]	Subtract B1 for each extra category quoted NOT These two categories are positive for all LAs This B1 can only be awarded if B1B1 already scored Good explanation of why these two are significantly different from the others. Must quote at least two figures from table, from any two categories, or two correct totals or averages
			Total	6	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
5	a		$k > 1.4$ $k < 0.25$ or $1.4 < k < \dots\dots$ $\dots\dots < k < 0.25$ (allow $k > 1.1$ to 1.6) (allow $k < 0.2$ to 0.3) (ranges as above) B1 B1	B1(AO2.2b) B1(AO2.2b) [2]	Allow $\geq$ and $\leq$ SC: $0.25 < k < 1.4$: B1B0 (ranges as on left) <u>Examiner's Comments</u> Centres are referred to the general comment on Question 11 under "Section A Overview" above. For the types of answers that were acceptable in Question 11 parts (b)(ii), (c) and (d), centres are referred to the published mark scheme. Few candidates answered this correctly. Most candidates did not understand which pairs of proportions to divide in order to find the limiting values of k. A helpful approach would be to draw a line from the origin, just meeting the point in a cluster nearest to the line $y = x$. This point provides the correct pair of proportions for one of the limiting values of k. Then repeat for the other cluster.	Allow "x"
	b	i	$0.797 > 0.5577$ or $-0.797 < -0.5577$ or $-0.797 > 0.5577$	B2(AO3.1b) (AO3.2a) [2]	$0.797 > 0.6055$ or $-0.797 < -0.6055$ B1 ± 0.5577 B1 <u>Examiner's Comments</u> Most candidates answered this correctly. Two common errors were "$-0.797 < 0.5577$, hence significant" and $-0.797 < -0.6055$ (i.e. a two-tail test).	Allow $\geq$ or $\leq$

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
		ii	There are clusters (or groups etc.) Apparent good correlation caused by clusters or Two clusters with no –ve corr'n within them or a comment similar to one of the above. AND Conclusion unreliable or Value of r is misleading oe	B1*(AO2.3) B1dep B1*(AO3.5b) [2]	or Not bivariate normal distribution B1 so use of tables for r not valid B1 Examiner's Comments Many candidates thought the statement was reliable, because the diagram does show negative correlation. Many other candidates said the conclusion was not reliable because the diagram omits data from a large number of LAs. A few noted the key point that the points on the diagram are in two clusters. But not many explained that this results in an apparent (but false) appearance of good correlation. A few candidates noted, correctly, that the points cannot be contained roughly within an ellipse and therefore the distribution was probably not bivariate normal, so the use of Pearson's coefficient is invalid.	NOT Too scattered Not represent whole pop Small sample Clusters not on reg line B1B0
	c		High prop of 65+ or Low prop of 18-24 Prop of young very similar, or ≈ 0.06 Proportion of senior to young is high	B1(AO2.2b) [1]	If consider only <u>one</u> age-group, must be proportion not number If consider <u>both</u> age-groups, allow eg Higher number of seniors than young or Many seniors, few young Examiner's Comments Many candidates gave good answers. Some, incorrectly, stated that the diagram showed that relevant LAs had a large <u>number</u> of older residents.	NOT: Similar proportions of 65+ Population is elderly

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	d		Top left points contain high prop of 18-24s. (So these LAs may be areas where there are universities or where they can recruit)	B1(AO2.2b) [1]	Shows places where large nos of 18-24s Shows where to focus recruiting. So universities can recruit. 18-24s are their target group. No need to specify "Top left group" Allow "students" or "young" instead of "18-24s" Any implication that diagram enables you to see information about location of young people <u>Examiner's Comments</u> There was a large variety of answers. Perhaps the best answers referred to the usefulness of the diagram in choosing areas where a university might concentrate its advertising. Any answer was accepted that implied that the diagram enabled members of the audience to identify areas where there was a high proportion of young people.
			Total	8	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
6	a		LA4 Large increase in numbers travelling by tram etc or Increase in tram etc greater than the others or There was an increase from 65 to 2495	B1*(AO2.2b) B1(AO2.2b) [2]	Not eg "LA4 or LA1", but allow "LA4 and possibly LA1, but less likely" Dep 1st B1* Allow any description of "tram etc" No need to mention "tram" if clear <u>Examiner's Comments</u> Most candidates saw the point and answered well. Some chose LA1 with a reason for which there is only a small amount of evidence in the data. For example: "LA1, because the number using underground, etc. increased by more than 8000 and the number driving decreased," or "LA1 because the increase in numbers using tram, etc. was the greatest out of the four LAs". Answers such as these were given no credit because they ignore the changes in total population and, more especially, the evidence for LA4 is overwhelmingly much stronger. The change in proportion is the one which matters most in this context, rather than the absolute change in number.

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Question			Answer/Indicative content	Marks	Guidance	
	b		Ratio for bus is approx same as ratio for all people or Recognition that total number has increased	B1*(AO2.2a)	Must include comment on total number Must refer to data in table	Not Bus growth is greater than other forms of transport
			No reason to suggest new bus routes	B1(AO2.3) [2]	Dep 1st B1*	Ignore all else. Eg "True but possibly not true": ignore the first part.
<u>Examiner's Comments</u> This question required more subtlety than part (a). Many candidates only considered the absolute increase in numbers using bus, etc. in LA1 and these candidates gained no marks. The key point was that there was also an increase in the total number of employees, and that this by itself could have accounted for the increase in bus usage. (In fact the percentage increases for all employees and for bus, etc. are almost identical, although candidates were not required to give this much detail.) This fact means that there is little evidence to support Julian's suggestion. Many candidates answered the question as if it concerned simply the increase in bus usage. In fact the question is not about increased bus usage, but is about the possibility of new bus routes having been introduced.						

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Question			Answer/Indicative content	Marks	Guidance	
	c		LA1, because: decrease in number driving cars despite increase in total number of people or all other LAs had increased numbers driving or ≥ 2 LA1 public transport categories increased	B1(AO2. 2b) B1(AO2. 2b) [2]	Must state LA1, but no mark for this yet eg LA1 is the only LA with decrease in driving: B1B1	Ignore all else
					<u>Examiner's Comments</u> Most candidates correctly identified LA1, but in their justification many candidates referred only to the decrease in numbers driving. These scored 1 mark. To gain the other mark candidates needed to give a comment adding further support to their choice of LA1. For example, they could note that the decrease in driving numbers happened despite an increase in the total number of employees in LA1. Or they could note that the decrease in numbers driving in LA1 was unique - none of the other three LAs showed a similar decrease. Or they could note that, alongside the decrease in driving numbers in LA1, there was an increase in numbers using each of tram, train and bus.	
			Total	6		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
7		a	Wrong total	E1 (AO 3.2b) [1]		
		b	$2046.3 \times 347 \div 348$ $= 2040.4$	M1 (AO 1.1a) A1 (AO 1.1) [2]		
		c	<u>2011</u> $2860.8 \times 348 \div 26526336$ $= 0.0375$ Hence 2011 higher <u>2001</u> $2040.4 \times 348 \div 23627753$ $= 0.0301$	M1 (AO 2.4) A1 (AO 2.2a) [2]	M1 both calculations A1 both results and conclusion	or without "× 348" $= 0.0000789 = 0.000086803$
		d	More metros built.	E1 (AO 2.2b) [1]	or sensible alternative	
			Total	6		
8		a	E.g. C is a London borough because it has a (very) large percentage of metro	E1 (AO2.2b) [1]		
		b	E.g. D is a metropolitan borough because it has a large percentage of metro E.g. A is a metropolitan borough because it has a large percentage of metro train	E1 (AO2.2b) E1 (AO2.2b) [2]	or bus or bus	
		c	E.g. D has some sort of metro, but A does not	E1FT (AO2.2b) [1]	or A has more trains (or buses) than D.	FT their two LAs from (ii)
		d	E.g. It looks like there is very little public transport available because a very small percentage of people use bus or train.	E2 (AO2.2b) (AO2.2b) [2]	E1 for very little available E1 for reference to supporting evidence	
			Total	6		

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance	
9		a	In all parts, once mark gained, ignore all else Advantage: Type 1 answers: State or imply compare proportions (or distributions or structure or profile or pattern) Examples: Can comp proportions (or distributions or structure or profile) Allow can see props Can compare areas' age groups relative to size of area Easier to see age group distributions Disadvantage: Type 1 answers: State or imply pop sizes not <u>easy</u> to compare Examples: Diag does not show relative sizes of the authorities R'd appears to have more in 0-17, but actually L'I has more in this group Hard to compare because diff nos rep by same size on diags Can't compare numbers (or results or pops or sizes) easily Can't compare numbers (or results or pops or sizes) without calculation	E1 (AO1.1) E1 (AO1.1) [2]	Allow eg "Group 1" for 0-17s etc. Advantage: Type 2 answers: State or imply with same scale, sizes of diags wd be very different Examples: Prevents diag from becoming too big or too small to use effectively If one set of values is a lot lower than the other, it will be hard to compare them on the same scale. Disadvantage: Type 2 answers: State or imply mismatch between diag size and pop size Examples: Confusing because same size diag for diff size populations Looks as if same no. of people in each Might miss the fact that scales are diff, looks as if more 0-17s in R'd than L'I	Allow "children" for 0-17s NOT e.g: Easy to compare large area with small Easier to see results Easy to compare populations Because L is bigger than R Can compare age in small & large areas NOT eg Can't compare results Can't compare numbers Easy to be mistaken when comparing

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					<u>Examiner's Comments</u> Many candidates gave correct answers here, although frequently they used far more words than were required. Some gave inadequate answers such as (for the advantage) "It's easier to see the results", "It's easier to compare the populations" or "Can compare ages in small and large populations". For the disadvantage some inadequate answers were "Can't compare results"; "Easy to make a mistake". Many candidates wrote long essays which did not necessarily gain any marks. Examples of brief, acceptable answers are: Advantage: The different scales make it easy to compare the age group proportions in Liverpool with those in Rutland. Disadvantage: The different scales make it hard to compare the age group numbers in Liverpool with those in Rutland.
		b	90000. Allow between 75000 & 95000 incl.	B1 (AO2.2b) [1]	Allow reasonable ans given as range eg "Much more than 50000 but < 100000" <u>Examiner's Comments</u> Most candidates answered this question correctly. A few wrote 100 000 even though the dot is clearly below this value.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		c	"L" = Liverpool. "R" = Rutland NB: Must be about 60-74s and/or 18-29s and/or 0-17s Answer type 1 Compare <u>proportions</u> in two age groups. Examples: Any two of eg: L has smaller prop of 60-74 (than R'd) L has smaller prop of 0-17s (than R'd) L has larger prop of 18-29s (than R) eg, L prop of 18-29s is 4 × R prop 18-29s R has smaller prop of 18-29s R has hier prop of 0-17s	E1 (AO2.2b) E1 (AO2.2b) [2]	Answer type 2 Compare gps with largest (or smallest) props. Allow "number" instead of prop only for this type of answer Examples: L's hiest no. (or mode) is 18-29s AND R's hiest no. (or mode) is 0-17s E1 only L's smallest is 75+ AND R's smallest is 18-29 E1 only (75+ allowed in this case only) NOT "number" except in ans about modes or smallest. Ignore all else. <u>Examiner's Comments</u> Again, many answers were inadequate, such as "Rutland has a higher proportion of middle aged people". Answers that compared numbers (as opposed to proportions) of people in the two areas were not accepted. For example, "Liverpool has more people in the 18-29 age group than Rutland.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		d	Must state gp who are likely to have babies ie 18-29s or 30-44s or 18-44s. (Allow 0-29s or "young") This gp is large in L, AND is small in R	E1 ind (AO2.4) E1dep (AO2.4) [2]	Inadequate ans: L high prop of young, who will have babies E1 R high prop of old E0 Allow "number" instead of "proportion" NOT just This gp is larger in L <u>Examiner's Comments</u> Most candidates recognised the key point here, which was that children are being born during the 10 years from 2011. However, some only stated that Liverpool had a large proportion of people in the potential child-bearing group, and failed to state that Rutland has a small proportion in this group. Some candidates just argued about the sizes of the groups changing as people get older, without considering new births at all. An example of an acceptable answer is: People in age group 18-29 (and/or 30-44) are likely to give birth to children during the next 10 years. There is a high proportion of this group in Liverpool, but a low proportion in Rutland. Acceptable answers to Question 13 parts (a), (c) and (d) can be found in the published mark scheme. Throughout this question, a common error was to write about "numbers", rather than "proportions", of people in the two areas.
			Total	7	